P04

Admittance and Conductance Formulations for Interconnected Grids

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Domain DOM_ENG

Abstract

Nodal admittance matrices in electrical transmission grids are weighted graph Laplacians with complex branch weights. Sparse Gaussian elimination and Kron reduction contract passive internal nodes.

Abstract quoted from J. Bergen et al., Power Systems Analysis (2000).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?